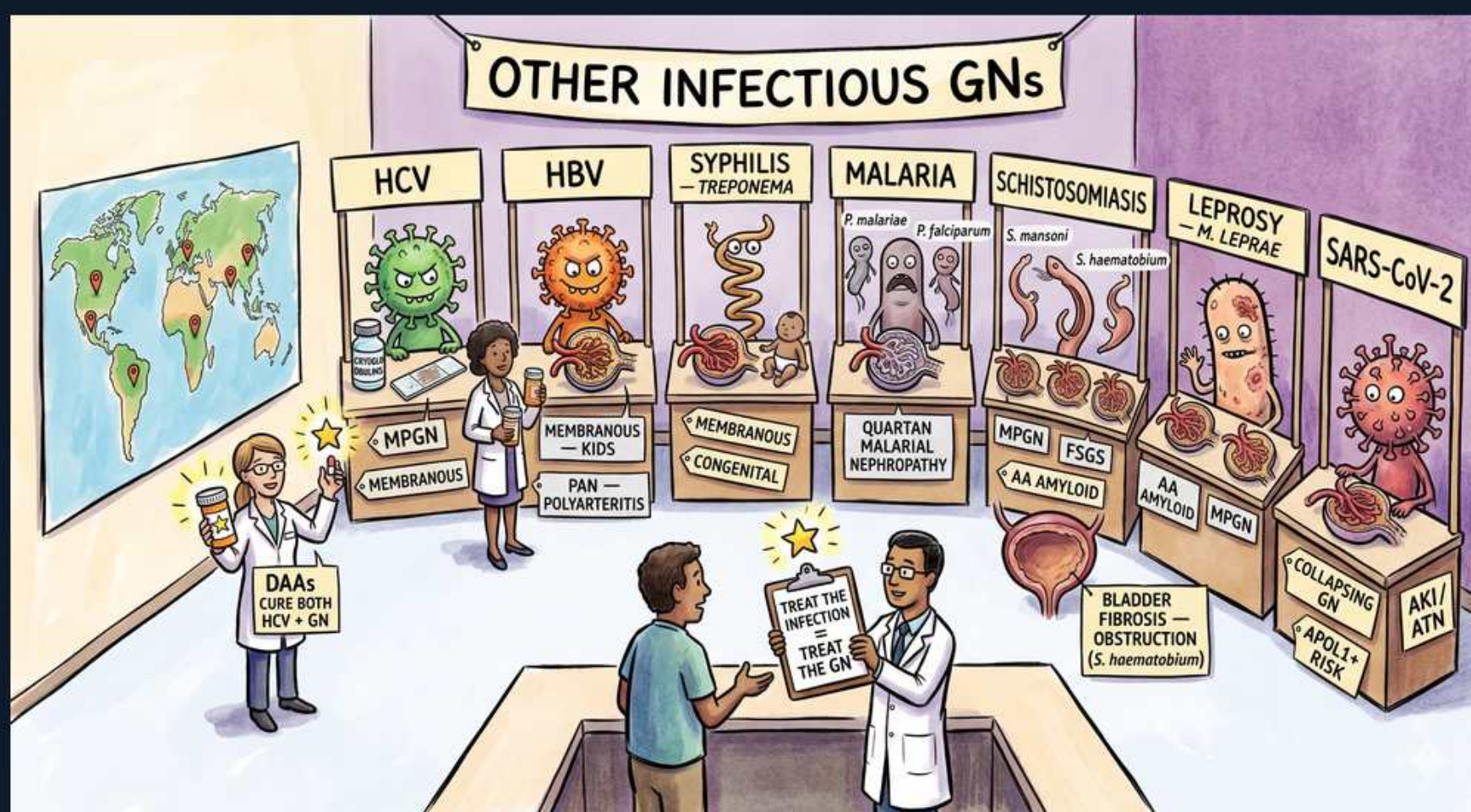


Infection-Associated Glomerulonephropathies — Other Infectious GNs



1. Other infectious GNs (overview)

WHAT YOU SEE

Banner: OTHER INFECTIOUS GNs over a museum of pathogen booths

WHAT IT ENCODES

Many chronic infections drive immune-complex glomerular disease through circulating antigen-antibody complexes. The unifying board theme: identify the infection, because treating the infection often treats the kidney.

BOARD TRAP

These are immune-complex GNs from chronic antigenemia — immunosuppression without treating the underlying infection can worsen the infection.

2. HCV → MPGN (+ cryoglobulins)

WHAT YOU SEE

HCV booth: MPGN, membranous; DAAs cure both HCV and GN

WHAT IT ENCODES

Hepatitis C classically causes type I membranoproliferative GN, usually with mixed (type II) cryoglobulinemia — low C4, RF positive, palpable purpura. Direct-acting antivirals (DAAs) that clear HCV can resolve the glomerulonephritis.

BOARD TRAP

Low C4 with cryoglobulins points to HCV-MPGN, not lupus; check HCV before reaching for immunosuppression — DAAs are first-line.

3. HBV → membranous (kids), PAN

WHAT YOU SEE

HBV booth: membranous in kids, pan-polyarteritis

WHAT IT ENCODES

Hepatitis B most classically causes membranous nephropathy, especially in children, and is the classic infectious trigger of polyarteritis nodosa (PAN). Antiviral therapy is the mainstay.

BOARD TRAP

HBV is membranous (and PAN); HCV is MPGN/cryoglobulinemic — a high-yield swap the boards love to test.

4. Syphilis → membranous, congenital

WHAT YOU SEE

Syphilis (Treponema) booth: membranous, congenital

WHAT IT ENCODES

Secondary and congenital syphilis can cause membranous nephropathy. It resolves with penicillin treatment of the infection.

BOARD TRAP

Membranous in a young patient with a rash/serologies — consider syphilis and treat the infection rather than starting immunosuppression.

5. Malaria → quartan nephropathy

WHAT YOU SEE

Malaria booth (P. malariae, P. falciparum): quartan malarial nephropathy

WHAT IT ENCODES

Plasmodium malariae causes quartan malarial nephropathy — a chronic immune-complex glomerulopathy that often progresses despite antimalarials. P. falciparum more often causes AKI/ATN from hemolysis and blackwater fever.

BOARD TRAP

P. malariae = chronic nephropathy (immune complexes); P. falciparum = acute kidney injury/ATN — don't conflate the species.

6. Schistosomiasis → MPGN/FSGS/AA

WHAT YOU SEE

Schistosomiasis booth (S. mansoni, S. haematobium): MPGN, FSGS, AA amyloid; bladder fibrosis

WHAT IT ENCODES

S. mansoni causes immune-complex GN (MPGN/FSGS) and AA amyloidosis. S. haematobium causes bladder/ureteral fibrosis with obstruction and is linked to squamous-cell bladder carcinoma.

BOARD TRAP

S. haematobium = urinary tract (bladder fibrosis, SCC), S. mansoni = glomerular/hepatic — the species map to different organ patterns.

7. Leprosy → AA amyloid, MPGN

WHAT YOU SEE

Leprosy (M. leprae) booth: AA amyloid, MPGN

WHAT IT ENCODES

Chronic lepromatous leprosy is a classic cause of secondary (AA) amyloidosis and immune-complex MPGN due to sustained inflammation.

BOARD TRAP

Chronic infection plus nephrotic-range proteinuria and apple-green Congo-red birefringence = AA amyloid, not primary GN.

8. SARS-CoV-2 → collapsing GN, AKI/ATN

WHAT YOU SEE

SARS-CoV-2 booth: collapsing GN, APOL1, AKI/ATN

WHAT IT ENCODES

COVID-19 can trigger collapsing glomerulopathy (COVAN) in APOL1 high-risk patients, plus AKI/ATN in severe illness. It mirrors the HIVAN collapsing pattern.

BOARD TRAP

Collapsing FSGS in an APOL1-risk patient after a viral illness points to COVAN/HIVAN-type injury, not classic primary FSGS.

9. Treat the infection = treat the GN

WHAT YOU SEE

Two clinicians holding signs: TREAT THE INFECTION / TREAT THE GN

WHAT IT ENCODES

The central rule across all infection-associated GNs: eradicating the underlying organism (antivirals, antibiotics, antiparasitics) is the primary treatment and often resolves the kidney disease. Immunosuppression is used cautiously and usually as an adjunct.

BOARD TRAP

Reaching for steroids/cyclophosphamide first can flare the infection — control the organism before or alongside any immunosuppression.

10. DAAs cure HCV + GN

WHAT YOU SEE

Clinician with star: DAAs cure both HCV + GN

WHAT IT ENCODES

Direct-acting antivirals achieve high HCV cure rates and, by clearing the antigen, can resolve HCV-associated cryoglobulinemic MPGN. Severe cryoglobulinemic disease may also need rituximab/plasmapheresis.

BOARD TRAP

DAAs treat the cause; rituximab/plasmapheresis are reserved for severe or rapidly progressive cryoglobulinemic disease, not routine first-line.

11. Geographic epidemiology

WHAT YOU SEE

World map with endemic hotspots highlighted

WHAT IT ENCODES

Travel and exposure history is a major board clue: malaria and schistosomiasis are tropical/endemic, leprosy in specific regions. Geography narrows the differential of an immune-complex GN.

BOARD TRAP

Don't anchor on autoimmune GN when the exposure history (travel, endemic region, freshwater) points to an infectious immune-complex cause.
